

Executive Summary: SQL Data Analysis

Project 3 | DecodeLabs Data Analytics Internship, Batch 2026 | Prepared by Peter Omotoso

1,200 orders analysed	\$1,264,761.96 total order value	\$1,053.97 average order value	41% orders Cancelled or Returned
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1. Goal

Use SQL queries to extract insights from a dataset, showing SELECT queries, WHERE, ORDER BY and GROUP BY, and the basic aggregations COUNT, SUM and AVG.

2. The Problem

A sales dataset of 1,200 orders (January 2023 to June 2025) held plenty of data but no answers. The business needed to know which products earn the most, how many orders actually reach the customer, which payment methods and marketing channels perform best, and how orders change over time.

3. What I Did

- Checked the data first: 1,200 rows, 14 columns and no NULL values, so COUNT, SUM and AVG are reliable.
- Wrote 14 SELECT queries on the order data.
- Filtered with WHERE (equality, number comparison, LIKE) and sorted with ORDER BY (ascending and descending).
- Grouped with GROUP BY and summarised with COUNT, SUM and AVG.
- Went beyond the minimum: HAVING to filter grouped results, and each product's percentage share of revenue.
- Followed the SQL execution order and avoided the alias trap (no alias used in WHERE).
- Checked every result against the totals (1,200 orders and \$1,264,761.96) before writing insights.

4. Results

- **Total order value:** \$1,264,761.96 from 1,200 orders. Average order: \$1,053.97.
- **Products are well balanced.** Chair (\$195,620.11) and Printer (\$195,612.61) lead with 15.5% of revenue each. Phone is lowest at \$151,722.39 (12.0%).
- **Fulfillment is the biggest concern.** Only 231 orders (about 19%) were Delivered. Cancelled (250) and Returned (247) orders make up 41% of all orders, and Cancelled orders alone hold \$276,396.21 in value.
- **Payments:** Credit Card has the highest average order (\$1,127.55); Debit Card the lowest (\$1,001.56).
- **Channels:** Instagram brings the most revenue (\$275,285.45); Referral the least (\$226,815.58).
- **Trend:** Orders fell from 510 in 2023 to 459 in 2024. The 2025 figure (231) covers January to June only.
- **Other:** Coupons were used on about 74% of orders. The 34 orders worth \$3,000 or more were all 4 to 5 units.

5. Recommendations

- Reduce cancellations and returns first, since this is where the most value is lost.
- Put more marketing effort into Instagram and Email, and look for ways to improve Referral.
- Track whether coupons increase order value, since most orders already use one.

6. Conclusion

This project shows SQL fundamentals: querying, filtering and grouping. Order value is balanced across products, but a large share of orders do not reach completion. That is the main opportunity for the business.

Deliverables: Project3_SQL_Data_Analysis.docx (14 queries with results and insights) and project_3_dataset_FIXED.xlsx (analysis workbook).